

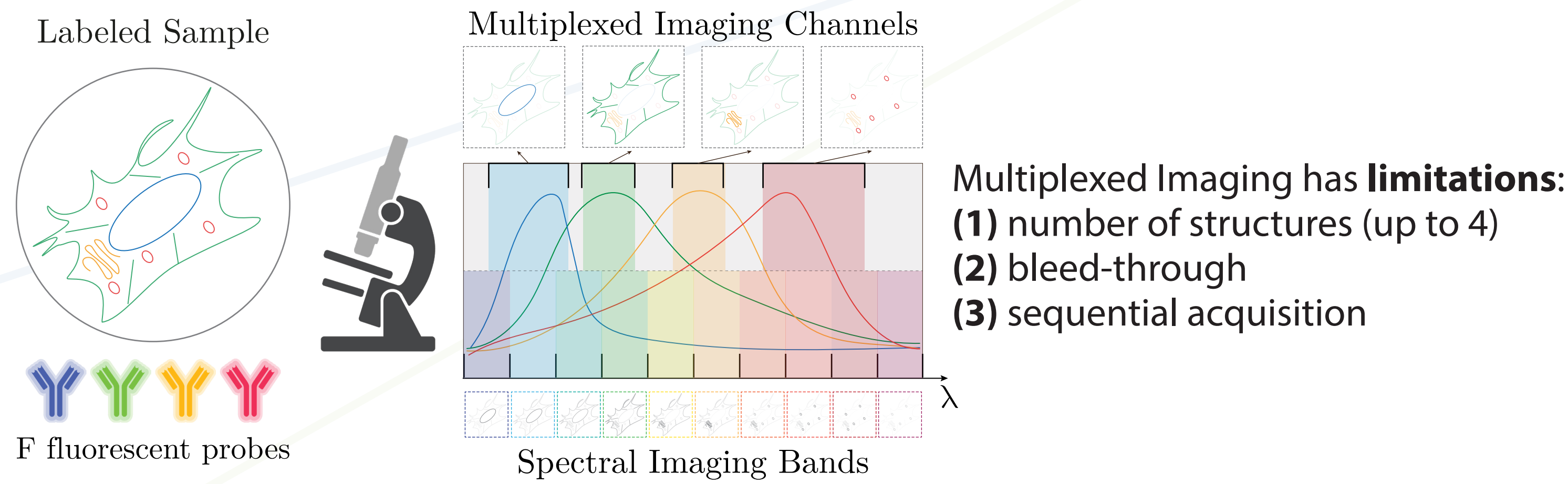
λSplit: Self-Supervised Content-Aware Spectral Unmixing for Fluorescence Microscopy

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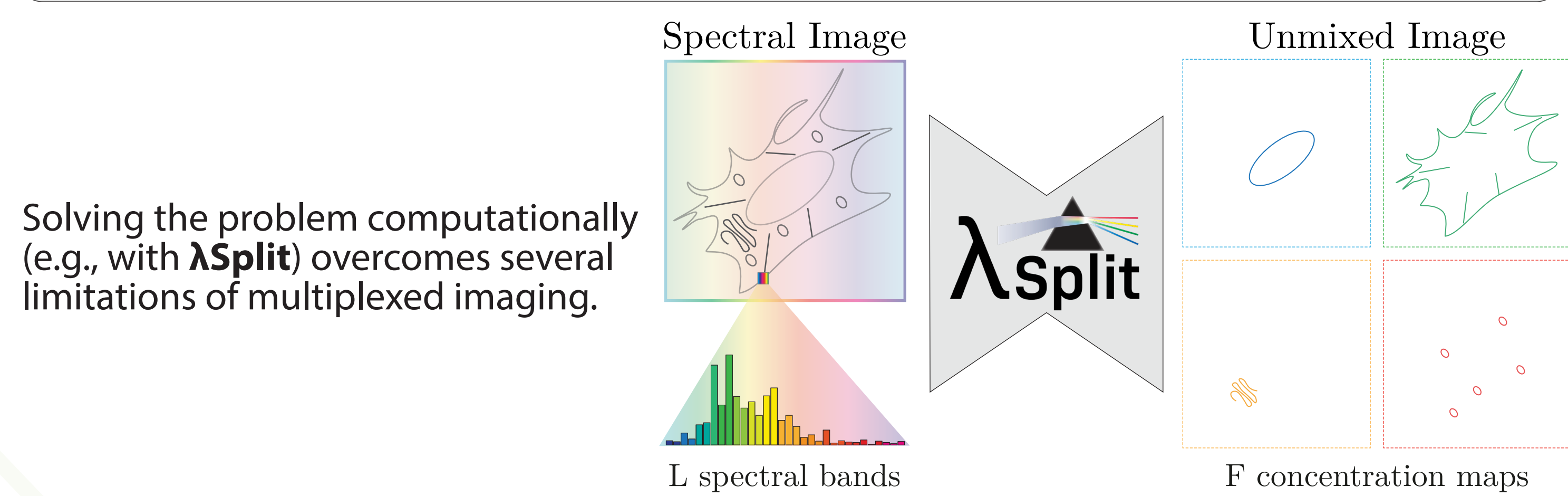
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1. Spectral Imaging and Unmixing

A biological sample labeled with **F Fluorescent Probes (FPs)**, whose emission spectra overlap, can be imaged using **Multiplexing** (band-pass filters to isolate FP contributions) or with **Spectral Imaging** (all unfiltered photons are acquired in L contiguous bands).

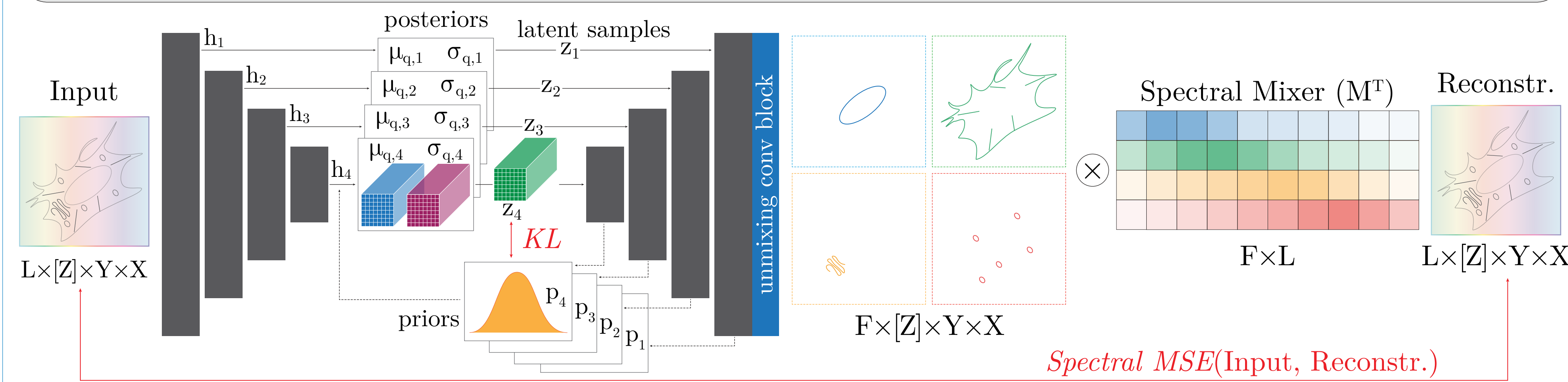


Spectral imaging shifts the burden from hardware-based to computational separation of FP emissions with **Spectral Unmixing**, the task of recovering FP concentration maps (**unmixed images**) from observed spectral images.



2. λSplit: a Self-Supervised, Physics-Informed LVAE for Spectral Unmixing

Classical Spectral Unmixing methods estimate FP concentrations **pixel-wise**, but are often unreliable under **high noise**, **spectral overlap**, or **low spectral dimensionality**, as the inverse problem is **ill-posed**, or even underdetermined ($L < F$). λSplit instead learns **structural priors** from spectral data, providing a **content-aware** solution that is robust even in these challenging regimes.

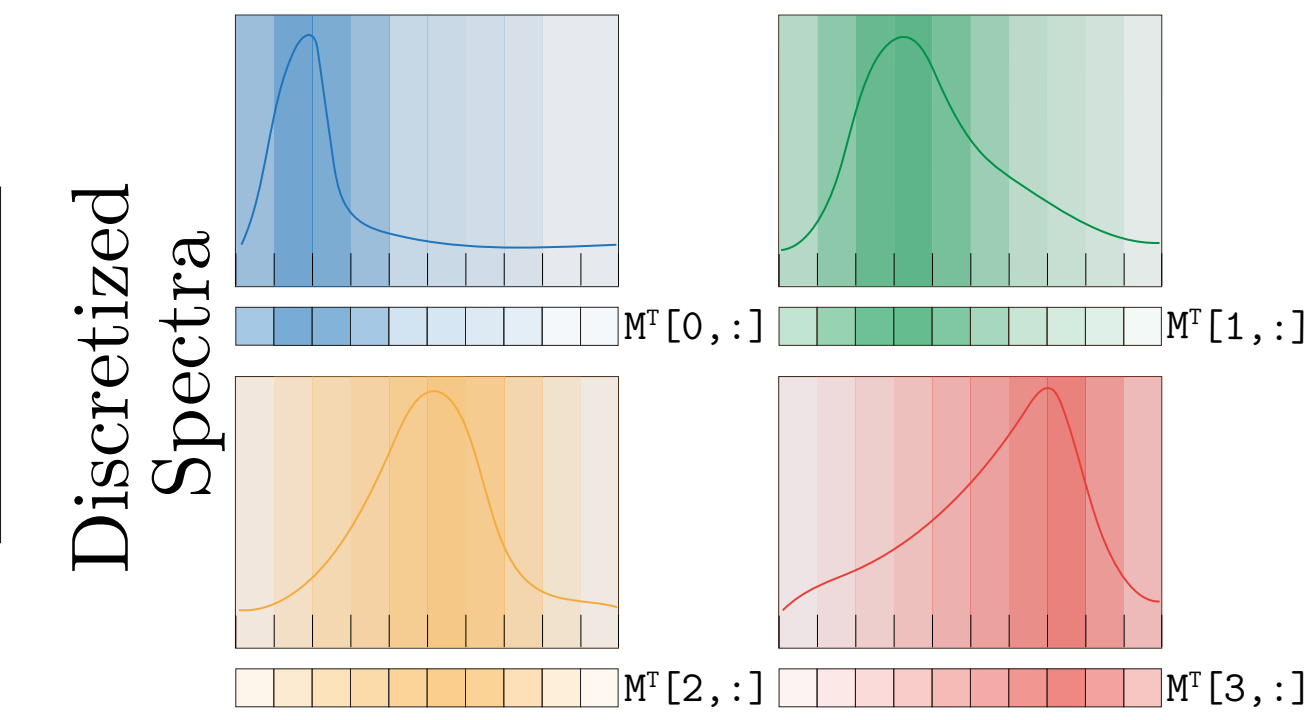


In fluorescence microscopy, spectral image formation is accurately described by a **linear mixing model**.

λSplit's fixed, but **differentiable Spectral Mixer** implements this model while enabling self-supervised learning (SSL).

$$S = MU + \epsilon$$

$$\begin{aligned} S &\in \mathbb{R}^{L \times [Z] \times Y \times X} \\ M &\in \mathbb{R}^{L \times F} \\ U &\in \mathbb{R}^{F \times [Z] \times Y \times X} \end{aligned}$$



Why an LVAE?

Hierarchical latents learn **multi-scale structural priors**

Implicit denoising via KL regularization and posterior samples averaging (MMSE)

Proven for image denoising and splitting in microscopy

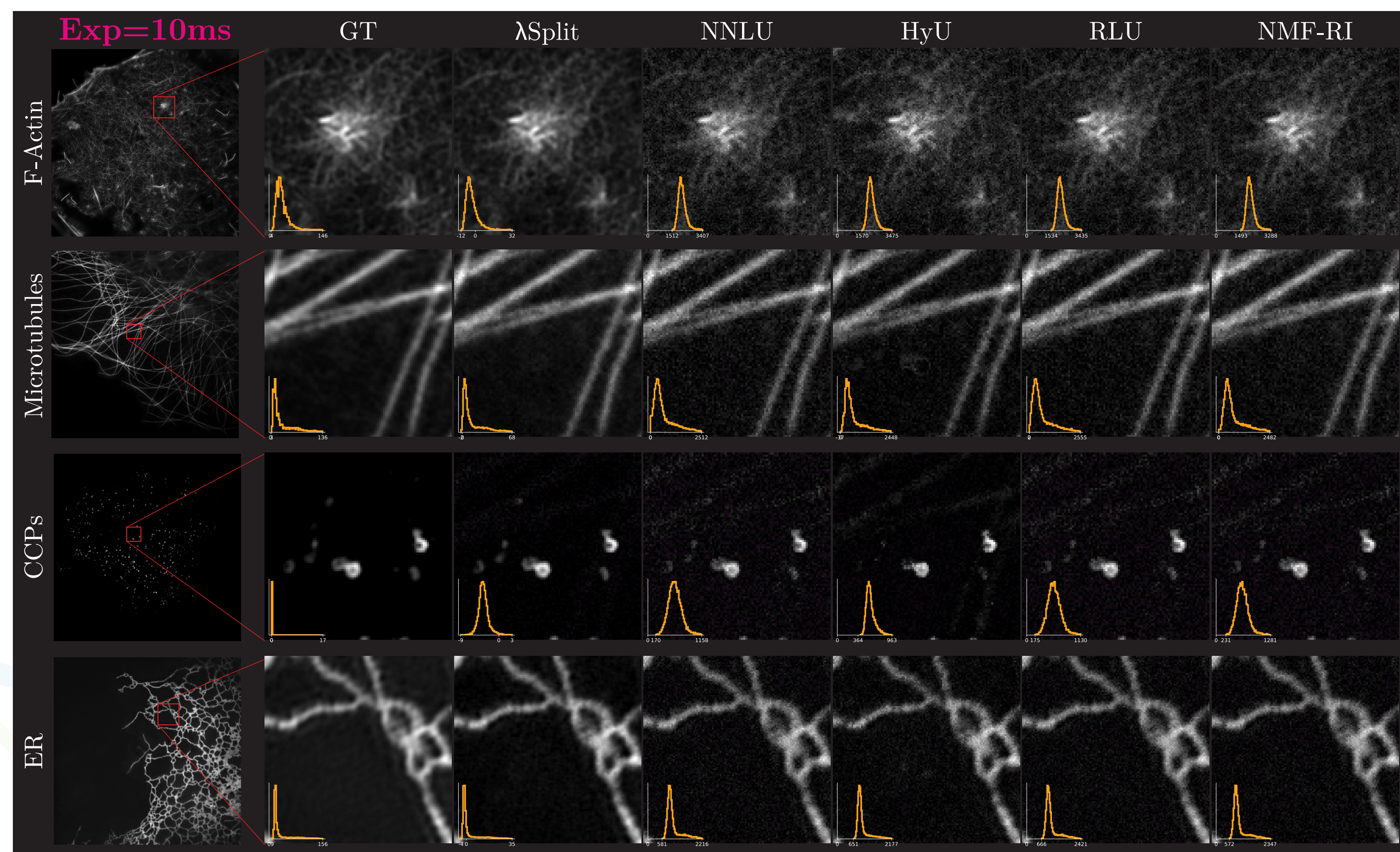
4. Spectral Unmixing & Increasing Noise

We simulate different levels of **pixel noise** by varying the **Poisson shot noise** (shorter detector **exposure** produces noisier images), while keeping the **Gaussian readout noise** fixed at a realistic value.

Experimental Setup: BioSR dataset, 4 structures with overlapping FPs acquired in 32 spectral bands.

Exp	SNR _{sp}	Method	PSNR ↑	MS3IM ↑	LPIPS ↓	SNR _u ↑
10ms 5.60		NNLU	30.42	0.923	0.503	10.69
		HyU	30.34	0.900	0.476	15.17
		RLU	30.54	0.925	0.502	11.45
		NMF-RI	30.56	0.925	0.497	10.59
		λSplit	38.55	0.981	0.138	56.71
20ms 9.77		NNLU	34.51	0.966	0.325	19.77
		HyU	33.29	0.940	0.318	28.83
		RLU	34.64	0.966	0.321	21.64
		NMF-RI	34.63	0.966	0.319	19.86
		λSplit	44.83	0.993	0.080	509.24
100ms 33.45		NNLU	42.76	0.989	0.072	71.88
		HyU	37.78	0.971	0.124	93.38
		RLU	43.00	0.989	0.068	74.90
		NMF-RI	42.98	0.989	0.071	73.25
		λSplit	46.15	0.994	0.087	660.22
300ms 57.43		NNLU	46.10	0.991	0.039	129.21
		HyU	39.12	0.976	0.092	162.10
		RLU	46.66	0.990	0.037	135.53
		NMF-RI	46.77	0.991	0.039	125.27
		λSplit	46.03	0.993	0.091	666.50

SNR_{sp} = SNR of spectral input SNR_u = SNR of unmixed pred. MS3IM = Multi-Scale SSIM



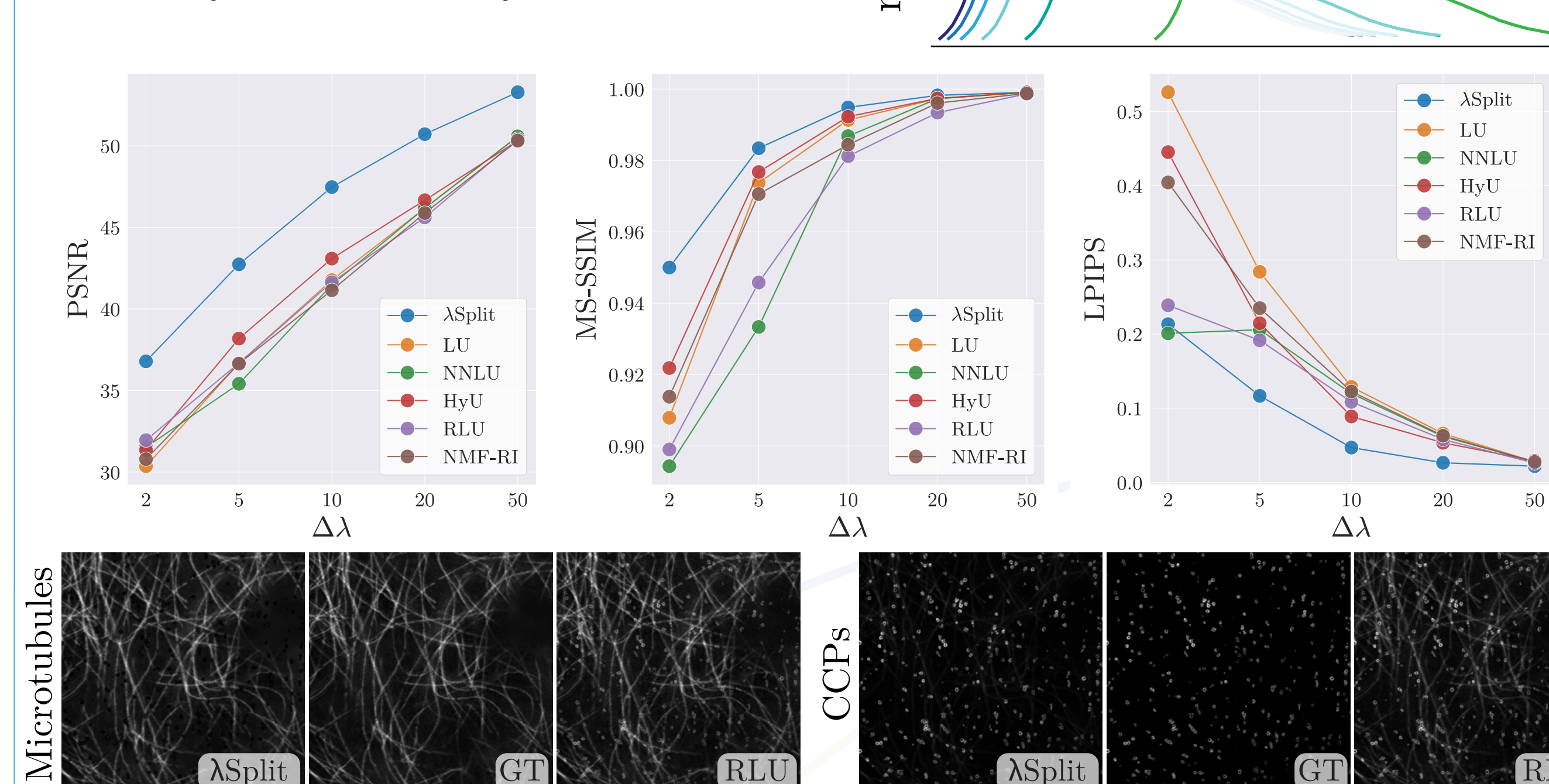
Qualitative comparisons at exposure = 10 ms show that pixel-wise methods **propagate the noise** from the input into the unmixed channels, while λSplit produces unmixed predictions that are simultaneously more accurate and denoised.

While classical unmixing baselines are competitive at high SNR, λSplit **consistently outperforms** them as noise increases, showing greater robustness in photon-limited set-

5. Spectral Unmixing & Increasing Spectral Overlap

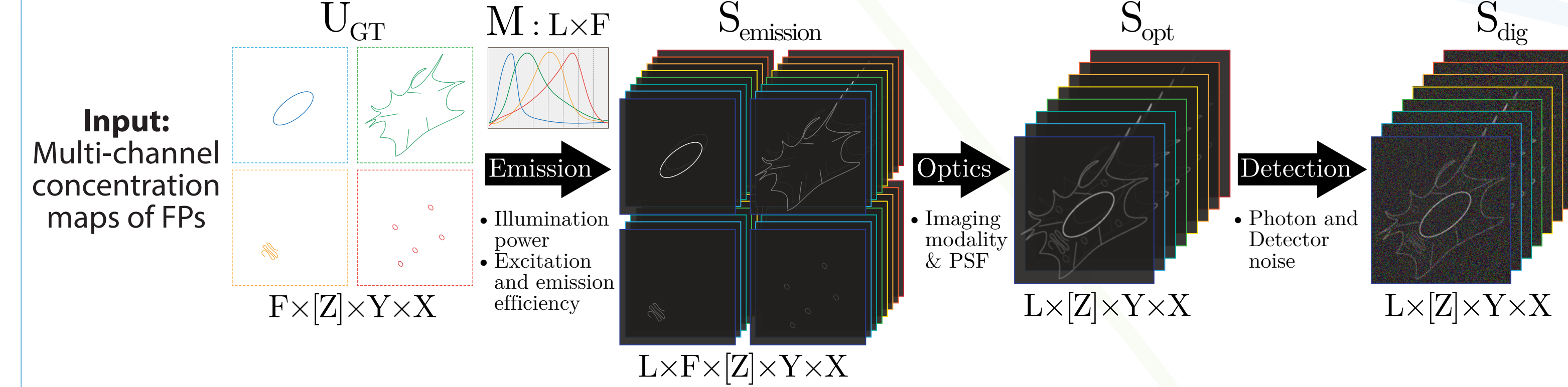
We simulate **increasing spectral overlap** by assigning the **same FP** to two different biological structures and rigidly shifting one spectrum by increasing $\Delta\lambda$. At high overlap, spectral unmixing is almost entirely based on **morphological discrimination** between structures.

Experimental Setup: Microtubule and CCP structures from the BioSR dataset acquired in 32 spectral bands, with shifts of $\Delta\lambda = \{2, 5, 10, 20, 50\}$ nm.



3. Data Simulation Pipeline, Datasets and Baselines

In microscopy, spectral data with paired GT unmixed images for quantitative evaluation are typically **unobtainable**. Hence, we rely on **synthetic datasets** generated using **microsim** [1].

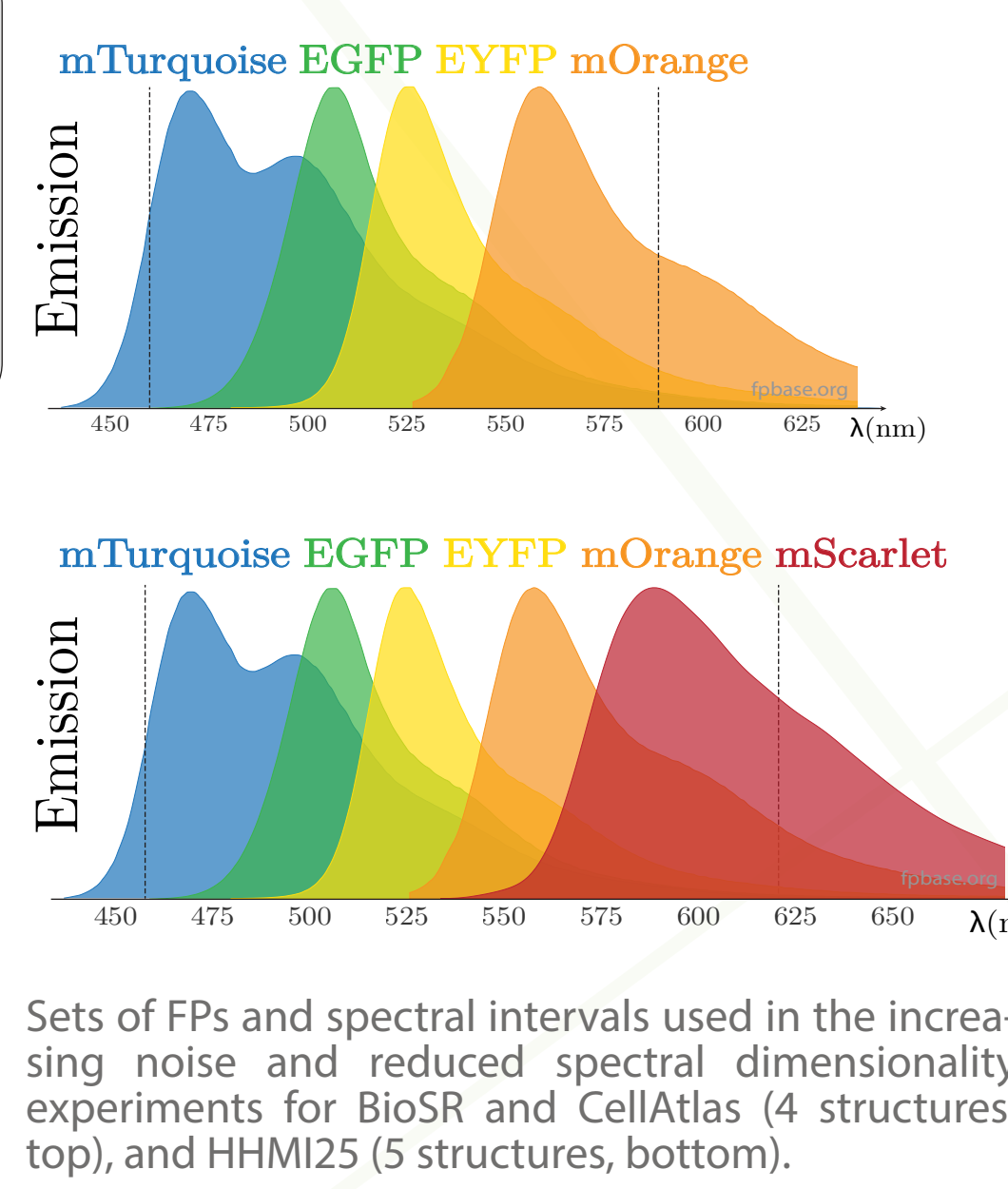


The data simulation pipeline enables controlled generation of realistic spectral images in **challenging unmixing** settings:

- I. **increasing noise**
- II. **increasing spectral overlap** of FP emissions
- III. **reduced spectral dimensionality** (fewer sp. bands)

As input to the simulation pipeline, we use three public datasets of multi-channel microscopy images: (I) **BioSR** [2] (2D); (II) **CellAtlas** [3] (2D); (III) **HMMI25** [4] (3D). We generate **58** benchmarks across different experimental settings.

We consider **10 classical** and **learning-based baselines**. Here, we only report the most competitive ones, namely: Non-negative Linear Unmixing (NNLU) [5], HyU [6], Richardson-Lucy Unmixing (RLU) [7], and Non-negative Matrix Factorization with Re-initialization (NMF-RI) [8].



6. Spectral Unmixing & Reduced Spectral Dimensionality

Experiment 1: Reducing # spectral bands keeping **per-band SNR constant**. This removes SNR as a confounder, isolating the numerical drawback of lower spectral dimensionality.

Experiment 2: Reducing # spectral bands keeping **total photon budget constant**. Fewer bands mean higher per-band SNR: we investigate whether this compensates for lower spectral dimensionality.

Experimental Setup: CellAtlas dataset, 4 structures with overlapping FPs.

Metric	Method	# bands				# bands			
		Same per-band SNR				Same total photon budget			
PSNR ↑	NNLU	26.21	27.30	28.56	36.02	26.79	28.36	28.56	25.83
	HyU	23.69	26.87	27.19	30.37	23.72	27.61	27.19	25.66
	RLU	26.39	27.56	28.91	36.68	27.13	28.63	28.91	26.15
	NMF-RI	26.31	27.76	28.38	36.49	26.77	28.70	28.38	26.13
	λSplit	27.25	29.95	31.12	36.46	27.45	31.54	31.12	31.32
MS3IM ↑	NNLU	0.863	0.916	0.938	0.986	0.874	0.934	0.938	0.865
	HyU	0.759	0.895	0.900	0.936	0.773	0.906	0.900	0.855
	RLU	0.866	0.921	0.943	0.988	0.882	0.938	0.943	0.872
	NMF-RI	0.857	0.918	0.934	0.989	0.866	0.935	0.934	0.872
	λSplit	0.878	0.960	0.971	0.988	0.868	0.967	0.971	0.971
LPIPS ↓	NNLU	0.239	0.267	0.232	0.128	0.192	0.238	0.232	0.392
	HyU	0.434	0.364	0.398	0.184	0.398	0.341	0.398	0.412
	RLU	0.238	0.258	0.220	0.125	0.187	0.229	0.220	0.387
	NMF-RI	0.237	0.247	0.227	0.124	0.185	0.217	0.227	0.389
	λSplit	0.283	0.267	0.313	0.155	0.273	0.238	0.313	0.218

Exp.1: λSplit's **learned structural priors** compensate for scarce spectral information.

Exp.2: λSplit's performance remains **stable across band configurations**, leading to improved versatility.

When the problem becomes **underdetermined** (3 bands), performance degrades. The structural prior alone cannot recover the missing spectral information.



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